WEEK 02

Comparisons and branches

ការប្រៀបធៀប និងមែកលក្ខខណ្ឌ

This week: learn, practise, build

សប្តាហ៍នេះ រៀន អនុវត្តនិងបង្កើត

01 · Python

Comparisons and branches
Understand the idea and trace examples.

02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

03 · Project

Review-first dispatch decision
Build, discuss and test with a peer.

One weekly component moves the final project forward.

ផ្នែកមួយរាល់សប្តាហ៍ជួយឱ្យគម្រោងចុងក្រោយរីកចម្រើន។

Python foundations

មូលដ្ឋាន Python

Comparisons and branches
Concept → trace a small example → explain its behavior

ប្រៀបធៀបតម្លៃហើយតាមដានមែកលក្ខខណ្ឌដែលត្រូវជ្រើស។

Conditions choose which statements run

លក្ខខណ្ឌជ្រើសរើសថ្លែងការណ៍ដែលត្រូវដំណើរការ

Compare values

A comparison returns True or False. >= includes the boundary; > excludes it.

Combine conditions

and needs both conditions; or needs either. not reverses a Boolean.

Choose one branch

if / elif / else takes the first matching block. Indentation defines each block.

Understand the behavior before changing the example.

យល់ឥរិយាបថក្នុងមុនកែឧទាហរណ៍។

A condition chooses the next path

លក្ខខណ្ឌជ្រើសផ្លូវបន្ទាប់

1

Input

A Boolean or comparison

2

Branch

True: run this block
False: use the other block

3

Continue

Statements after the branch run for either path

Read the diagram, then trace the same steps in code.

អានដ្យាក្រាម រួចតាមដានជំហានដដែលក្នុងកូដ។

Syntax to read and reuse

វាក្យសម្ពន្ធសម្រាប់អាន និងប្រើឡើងវិញ

Python / syntax	Meaning	When to use it
a >= b	Compare; include equality	Check a boundary
not / and / or	Combine Boolean values	Express a condition
if / elif / else	First matching branch	Choose one outcome
: + indentation	Begin and group a block	Control what runs together

Python Basics · sections 00, 04 · use only when you need a lookup.

Python Basics ផ្នែក 00, 04៖ មើលយោងតែពេលចាំបាច់។

Trace one selected branch

តាមដានមែកលក្ខខណ្ឌដែលបានជ្រើស

```
01 score = 20
02 enabled = True
03 print(score >= 20)
04 if enabled and score >= 20:
05     print("ready")
06 else:
07     print("waiting")
08 print("finished")
```

1. Expected: True, ready, finished.
2. Try score = 19, then enabled = False.
3. The final statement runs after either branch.

Predict → execute → explain the value at each step.

ព្យាករ → ដំណើរការ → ពន្យល់តម្លៃនៅជំហានមួយៗ។

Comparisons produce Boolean values

ការប្រៀបធៀបផ្តល់តម្លៃBoolean

```
01 total_portions = 20
02 reviewed = True
03 print(reviewed)
04 print(total_portions >= 20)
05 print(total_portions > 20)
06 print(reviewed and total_portions >= 20)
```

1. Predict each Boolean result.
2. >= includes equality; > excludes it.
3. Use Boolean values without quotes.

A comparison evaluates to True or False.

ការប្រៀបធៀបផ្តល់ True ឬ False។

if / else: follow the selected block

if / else៖ តាមដានផ្លូវដែលបានជ្រើស

```
reviewed = False
if reviewed:
    next_step = "inspect workload"
    print("Review recorded")
else:
    next_step = "coordinator follow-up"
    print("No review recorded")
print(next_step)
assert next_step == "coordinator follow-up"
```

1. Follow the path for False.
2. Both branches assign next_step.
3. Indentation defines each block.

A shared print runs after either branch.

print មួដំណរីការក្រោយមែកលខ្ណាណាមួយ។

if / elif / else: first matching branch

if / elif / else៖ មែកដំបូងដែលត្រូលក្លែណ្ហ

```
containers = 4
if containers == 0:
    draft_label = "empty"
elif containers < 4:
    draft_label = "small"
else:
    draft_label = "several"
print("Draft:", draft_label)
assert draft_label == "several"
```

1. Which branch handles four containers?
2. Trace zero → empty; three → small
3. Only the first matching branch runs

Order matters: zero matches empty before the small-count comparison.

លំដាប់សំខាន់៖ សូន្យត្រូ empty មុនធៀបចំនួនតិច។

and / or: compare the truth values

and / or៖ ប្រៀបធៀបតម្លៃ និមិត្តកែ

```python
reviewed = False
total_portions = 21
print("AND:", reviewed and total_portions >= 20)
print("OR:", reviewed or total_portions >= 20)
assert (reviewed and total_portions >= 20) is False
assert (reviewed or total_portions >= 20) is True
```

1. Predict the two expressions.
2. False and True differs from False or True.
3. Change one input and compare again.

and requires both conditions; or requires at least one.

and ត្រូវការលក្ខខណ្ឌទាំងពីរ; or ត្រូវការយ៉ាងតិចមួយ។

Boundary testing: > versus >=

តេស្តព្រំដែន: > ធៀប >=

```
total_portions = 20
buggy = total_portions > 20
policy = total_portions >= 20
print("Buggy:", buggy)
print("Policy:", policy)
assert buggy is False
assert policy is True
```

1. Which comparison agrees with the stated policy?
2. At 19 both comparisons are False
3. At 21 both comparisons are True

Keep the agreed expectation while diagnosing the implementation.

រក្សាលទ្ធលទ្ធផលដែលបានព្រមព្រៀង ពេលរកកំហុសក្នុង។

Hands-on practice

ឈុតសកម្មភាព

Week_02.ipynb · Session 2
Predict → run → change → explain

ព្យាករណ៍ ដំណើរការ ផ្លាស់ប្តូរ និងពន្យល់លទ្ធផល។

Your turn: practise the same Python skills

ដល់វេនអ្នក៖ អនុវត្តជំនាញ Python ដដែល

Compare

Try 19, 20 and 21.
Predict > 20 and >= 20.
Explain the equality case.

Trace a branch

Toggle reviewed.
Follow the indented block.
Locate the shared print.

Combine

Compare and with or.
Use reviewed = False.
Try a total above 20.

Use the matching exercises in this weekly notebook. Explain what changed and why.

ប្រើលំហាត់ត្រូវគ្នានៅក្នុង notebook សប្តាហ៍នេះ។ ពន្យល់អ្វីដែលបានផ្លាស់ប្តូរ និងមូលហេតុ។

Check your work before asking for help

ធ្វើការងារមុនសុំជំនួយ

Predict first

Write what you expect before executing.

Change one thing

Compare values, types, paths or returned results.

Explain the cause

Use the relevant Python rule. Retest your repair.

Trace comparisons, Boolean expressions and the selected branch.

តាមដានការប្រៀបធៀប កន្សោម Boolean និងមែកដែលជ្រើស។

Project studio

វគ្គនៃវគ្គម្រោង

Review-first dispatch decision
Short design briefing → most time for your work

ធ្វើការត្រួតពិនិត្យមុនផ្ញើស hold, standard ឬ team។

The people’s task this week

កិច្ចការរបស់អ្នកច្រើនាសប្តាហ៍នេះ

1

Someone needs help

The coordinator needs a rule for when a request can proceed.

2

This week, you will

Use the supplied review decision and total to choose hold, standard or team.

3

A useful result

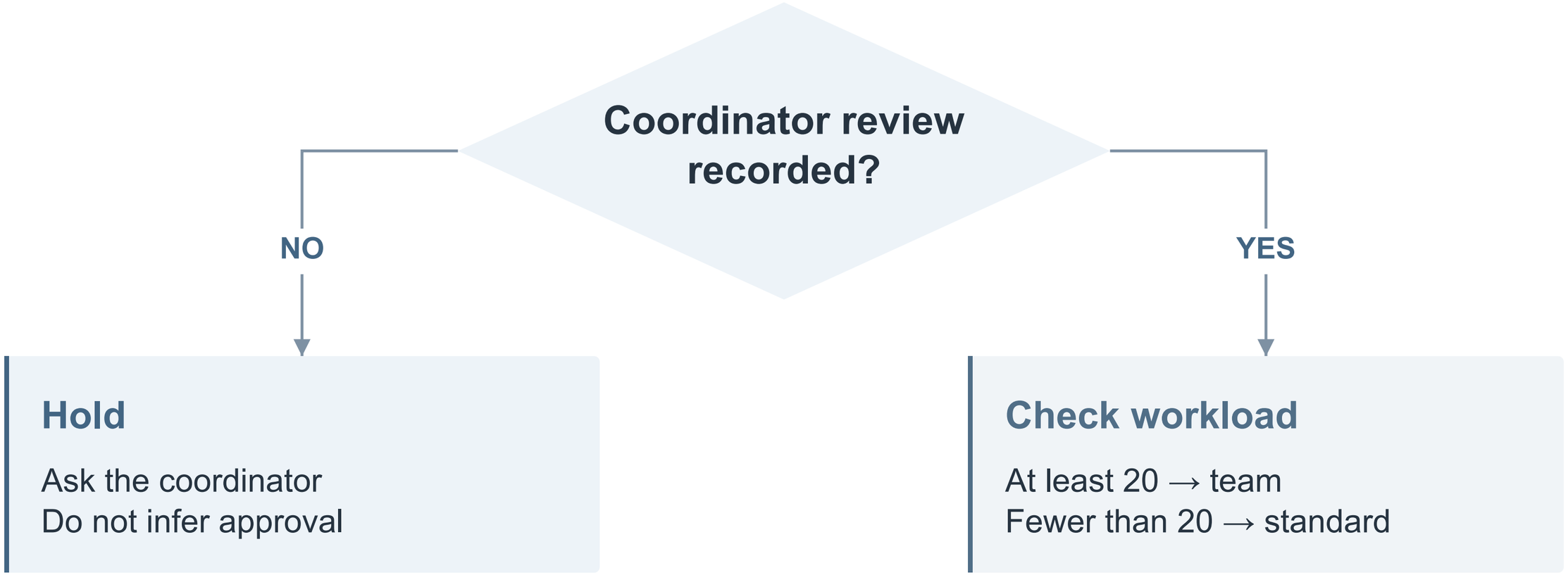
The coordinator can see the label and the reason for it.

Use the supplied starting files. Focus on one useful result and check it.

ប្រើកសារចាប់ផ្តើមដែលផ្តល់។ ផ្តោតលើលទ្ធផលមានប្រយោជន៍មួយ ហើយធ្វើតេស្ត។

Sketch the rule before implementing it

គូសច្បាប់សង្ខេបមុនសរសេរកូដ



Review comes before the workload decision.

ត្រួតពិនិត្យមុនសម្រេចតាមបន្ទុកការងារ។

Your implementation TODO

ការងារដែលអ្នកត្រូវសរសេរក្រូ

Supplied starting point

The notebook provides test inputs and a branching starter.

You implement / do

Implement review-first branching: unreviewed → hold; reviewed and total >= 20 → team; otherwise standard.

Check the evidence

Test 19, 20 and 21 with both review flags. Unreviewed always stays hold; exactly 20 is team when reviewed.

Unreviewed stays hold; reviewed totals of at least 20 use team.

មិនបានត្រួតពិនិត្យ hold; បានត្រួតពិនិត្យ និងសរុប ≥20៖ team។

Show the component working

បង្ហាញថាផ្នែកនេះដំណើរការ

Step	What to show
1 · Locate	Identify the input, this component and its next user.
2 · Implement	Implement hold / standard / team with review first.
3 · Check	Test 19, 20, 21 with review True and False.
4 · Explain	Show your code, its actual result and why it behaves that way.

Unreviewed stays hold; reviewed totals of at least 20 use team.

មិនបានត្រួតពិនិត្យ hold; បានត្រួតពិនិត្យ និងសរុប ≥20 team។

Further reading in English

ឯកសារអានបន្ថែមជាភាសាអង់គ្លេស

1 Python tutorial: if statements

https://docs.python.org/3/tutorial/controlflow.html#if-statements

2 Python Boolean operations

https://docs.python.org/3/library/stdtypes.html#boolean-operations-and-or-not

Read one relevant section, then bring an example and a question.

អានផ្នែកដែលពាក់ព័ន្ធមួយ យកឧទាហរណ៍មួយ និងសំណួរមួយមកជាមួយ។